

Embeddings

Positional Encoding $\rightarrow$

Parallel Computation $\leftarrow$ Parallely

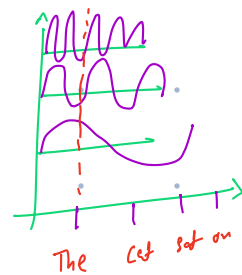
RNN

$L > 1000$ GPUs

1-1sec

$L > 1000$ /sec

$\downarrow$
(The cat sat on Mat) $\rightarrow$ Sinusoidal Waves



Positional Encoding + Embedding

[1, 1.5, 1.7, 1.8]

[2, 2, 3, 4.5]

3, 3.5, 4.7, 7.8

Values $\leftarrow$

Attention Mechanism $\rightarrow$

$O(n) \rightarrow O(1)$

The cat - - - Sat

10 - 0.99

No long Range

The [cat] - - - [Sat]

Self Attention

A B C $\rightarrow$ ①
/ / /
B,C A,C A,B $\rightarrow$ ②

A = B-70%, C-30%

Query - Z
Q (LLM)

(k)
A - ML
B - LLM
C - DM

$\rightarrow$ Value

Attention Score
 $Q, K, V \leftarrow$
/ / $\rightarrow$ Value
Query Key

① The cat sat - Input

↳ Tokenization

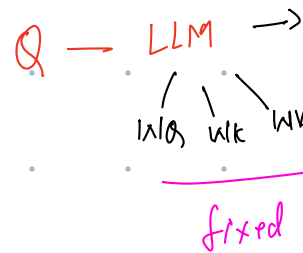
↳ Embeddings

↳ Q, K, V

↳ W_Q
 W_K
 W_V

} Randomly
↳ Trained

↳ fixed - Inference



② Generate Query, Key and Value

$$Q = X \cdot W_Q$$

$$K = X \cdot W_K$$

$$V = X \cdot W_V$$

↳ Compute Query with Every key

$$\text{L} \rightarrow Q K^T$$

↳ Softmax - $\checkmark$

↳ Attention Weights

→ Multiply V → New Contextual Embedding

$$d_{\text{model}} = 1536 \rightarrow 12000$$

$$\text{No of Heads} = 12$$

$$W = d_{\text{model}} \times d_k$$

$$d_k = \frac{1536}{12} = 128$$

$$d_k = 128$$

→

$$W_Q = 1536 \times 128$$

$$W_K = 1536 \times 128$$

$$W_V = 1536 \times 128$$

Single Attention Head →

W_K
 W_Q
 W_V

The $\rightarrow \underline{X} \times \underline{W_q}$
 Cat $\rightarrow \underline{X} \times \underline{W_q}$
 Sat $\rightarrow \underline{X} \times \underline{W_q}$

$$10 = \begin{bmatrix} 1536 \\ \times \\ 10 \end{bmatrix}$$

10 Tokens
 1536 - Dim
 $d_k = 128$

$$\begin{aligned} X &= (10 \times 1536) \\ Q &= X \cdot W_q \\ &= (10 \times \cancel{1536}) \times (\cancel{1536} \times 128) \\ &= (10 \times 128) \end{aligned}$$

$$\begin{aligned} K &= X \cdot W_k \\ &= (10 \times \cancel{1536}) \times (\cancel{1536} \times 128) \\ &= (10 \times 128) \end{aligned}$$

$$V = 10 \times 128$$

$\left. \begin{matrix} Q \\ K \\ V \end{matrix} \right\} (10 \times 128)$

$$Q \cdot K^T$$

$$\begin{aligned} &= (10 \times \cancel{128}) \times (\cancel{128} \times 10) \\ &= (10 \times 10) \rightarrow \text{Attention Matrix} \end{aligned}$$

	The	cat	sat
The	—	—	—
cat			
sat			

Attention to Values

$$\begin{aligned} V &= (10 \times 128) \\ \text{Attention} &= 10 \times 10 \end{aligned}$$

$$\begin{aligned} &\Rightarrow (10 \times 10) \times (10 \times 128) \\ &= (10 \times 128) \\ &\rightarrow \text{Attention output / Head} \end{aligned}$$

10 - 10x10 L100
 1000 - 1000x1000 L1M
 = 10 lakhs L>1M

10,000 $\rightarrow$ 10,000 x 10,000
 $\rightarrow$ 100 Million

12 Heads

Model = 1536

$$\rightarrow 10 \times 128$$

$$- N_0 = 12$$

$$10 \times 128$$

$$+$$

$$12$$

$$= 10 \times 1536 \rightarrow \text{Model Dimension}$$

$$\Rightarrow 1536 \times 128$$

$$= 196608 \text{ parameters}$$

$$\rightarrow \begin{matrix} 128 \\ 128 \\ 128 \end{matrix}$$

$$= 196608 \times 3$$

$$= 589824$$

$$= \underline{590K \text{ parameters}}$$

Self Attention Equation

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

$\rightarrow$ softmax $\rightarrow$ Deep

$$\sqrt{d_k}$$

$$= [1, 2, 3]$$

$$\hookrightarrow [0.09, 0.24, 0.70]$$

$$[100, 110, 120]$$

$$\hookrightarrow [0.00, 0.00, 1.00] \rightarrow \text{Meaning}$$

$$\sqrt{d_k} = \sqrt{128} = 11.31$$

$$\text{score} = [18, 22, 25]$$

$$\text{soft} = [0.001, 0.05, 0.949]$$

$$\sqrt{dk} = 11.31$$

$$= 18/11.31 = 0.80$$

$$= 22/11.31 = 0.97$$

$$= 25/11.31 = 1.11$$

$$\rightarrow (0.28, 0.33, 0.39)$$

$\hookrightarrow$ Attention

1) Sparse Attention
 $\hookrightarrow$ BigBird
 $\hookrightarrow$ Longformer

2) Sliding window Attention
 $(\text{window} = 2)$

34 (5) 67 $\rightarrow$ Mistral

3) Global + Local
 $\hookrightarrow$ Token

4) Linear Attention

$$QK^T$$

$$\hookrightarrow (QK^T)V$$

$$K^TV \downarrow Q(K^TV)$$

$$O(n^2) \downarrow O(n)$$

5) Flash Attention
 $\hookrightarrow O(n^2)$

$\hookrightarrow$ GPU Attention matrix

CPU Residual

GPT
 Llama
 GPT

Residual connection

$\rightarrow$ Multi-Head + input



Layer 1 - Layer 2 - Layer 3 ... Layer 48

↳ Information - disappears

↳ Gradient

↳ Training
difficult

Residual

(Layer 1) → Layer 48

↳ Gradient

↳ preserve information

↳ Deep models

→ Identity Mapping

learning →

output = Input

Layer Normalization

↳ Attention → Value large
small

The [200, 4, 5000, -20] > Scales

cat [0.2, 0.01, 0.01, 3]

Neural Network → inconsistent
scale

↳ Normalise

[5, 10, 15]

$$\text{Mean} = (5 + 10 + 15) / 3 = 10$$

Parameters - LN

- Gamma

- Beta

= Subtract this mean

$$= [-5, 0, 5]$$

Variance =

$$SD = [-1, -1, -1] \rightarrow \text{similar stable}$$

FFN

↳ Mixed information between Token

↳ Heavy Computation - FFN

(Dog → Animal → Pet)

↳ Dog is Related to pet

Dog is friendly Animal
Domestic
Companion Animal

Rich Transformation } FFN

→ 512 → (2048 → Relu) → 512
FFN

256 × 256

↳ Dog

↳ limited

↓

Upscale

(2048 × 2048)

↳ Manipulate

GPT Model

↳ Increase Dimension

(4096 → 11000 → Activat → 4096)

↳ Rich Transformation

King → [1.1, 0.5, 3.2]

Complex

↳ [1.7, 0.6, 3.9] → pet

↳ Heavy - FFN

↳ final

- 8/11

(2.7, 2.8, 3.9) → Rich semantic
features

Encoder - Decoder → Translation
75-flan

Encoder → BERT

Decoder → GPT, Claude, Gemini

Parameters

↓
2.7 T
30B

Context window

↓
262K
1M }

1M



Gemini →
OpenAI
Claude

Deepseek →

Offline Market → Prem

Private Instance → Secure

2.8T - Kimi K3

↳ local Machine

Matrix

↳ Many
Mult.

16 cores

↳ 16 core

↳ Mod

↳ Mod

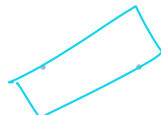
5-6-2

30 Billion

↳ 8bit - 1Byte

$30B \times 1\text{Byte}$
 $= 30\text{GB}$

RAM



16GB - 32GB

↳ 10GB

22GB

↳ 20GB

512GB

↳ Yes

Yes

No